University of Florida, Department of Mathematics

Topology PhD Exam, September 2002

Work the following problems and show all work. Support all statements to the best of your ability.

1. Use the Mayer–Vietoris sequence for reduced singular homology to compute $H_*(S^n)$.

2. Show that there is no retraction of the Möbius band onto its boundary.

3. Does there exist a covering space of the figure eight with nontrivial abelian fundamental group?

4. Show that the 2-dimensional sphere with three points deleted cannot be a topological group.

5. Let M be a 4-dimensional closed simply connected manifold. Show that every continuous map $f : M \to M$ which is homotopic to the identity has a fixed point.

Answer the following with complete definitions or statements or short proofs.

6. State the Seifert–van Kampen Theorem.

7. State the Lefschetz Fixed Point Theorem.

8. State the Jordan Curve Theorem.

9. State the Arzelà–Ascoli Theorem.

10. Let X be a finite complex. Find $\chi(X \times S^n)$.

11. Let $S_r(x, y)$ denote a circle of radius r centered at $(x, y) \in \mathbb{R}^2$. Let ΣR be the suspension over Hawaiian ring $R = \bigcup_{n=1}^{\infty} S_{1/n}(1/n, 0)$. Is ΣR

(a) locally path connected?

(b) locally simply connected?

(c) semi-locally simply connected?

12. State the Poincaré Duality Theorem.

13. Define retraction and deformation retraction.

14. State the Universal Coefficient formula for homology.

15. Let X be a complete metric space such that no point is isolated. Can X be countable?